SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

A) The large difference between the **training MSE (0.85)** and **testing MSE (5.20)** indicates that the neural network is **overfitting**.

Analysis:

- Training MSE = 0.85: The model fits the training data very well.
- Testing MSE = 5.20: The model performs poorly on unseen data.
- This means the model has learned the training data (including noise) instead of generalizing to new data.

Possible Reasons:

1. Overfitting due to a complex neural network.
2. Insufficient training data, making it difficult to generalize.
3. Noisy or irrelevant features in the dataset.
4. Too many training epochs, causing the model to memorize the data.
5. Lack of regularization techniques such as dropout or L2 regularization.

Corrective Measures:

- Use more training data or apply data augmentation (if applicable).
- Reduce model complexity by using fewer hidden layers or neurons.
- Apply regularization (Dropout, L1/L2 regularization).
- Use early stopping to stop training before overfitting occurs.

- Perform feature selection to remove irrelevant features.
- Use cross-validation to evaluate the model's generalization ability.

Conclusion:
The low training MSE and high testing MSE show that the model is **overfitting**. Using techniques such as **regularization, early stopping, reducing model complexity, and increasing training data** can improve the model's performance on unseen data.

Code of Conduct:

I K. Hema Sai Siddartha (192325036) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
K. Hema Sai Siddartha

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5)**	**Level 3 – Proficient (4)**	**Level 2 – Developing (2-3)**	**Level 1 – Beginning (0-1)**
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided